

AI Use Case Document

MailGuard — AI-Powered Email Spam Detection

Executive Summary

MailGuard is an AI spam detection system integrated into corporate email servers. It classifies incoming emails as spam or not-spam and routes them accordingly.

System Description

MailGuard uses a fine-tuned BERT classifier trained on a dataset of 10 million labeled emails. It analyzes email headers, sender reputation, content patterns, and link destinations to classify each message.

Users and Affected Persons

Users are IT administrators at private companies. Affected persons are employees whose incoming email is filtered. False positives are routed to a spam folder that users can review.

Outputs

Binary classification: spam or not-spam. Spam emails are moved to a quarantine folder. Users can whitelist senders and recover misclassified emails at any time.

Human Oversight

Full human override is available at all times. No decisions about persons are made. The system only routes emails — it does not block, penalize, or profile individuals.

Deployment Context

Deployed as an on-premise software package for corporate IT infrastructure. Used by private companies in the EU for internal email management.